MATH 301: Homework 2

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Question 1: Section 2.1-10

Part (a): If $a < b$ and $c \leq d$, prove that $(a + c) < (b + d)$

Suppose $a < b$ and $c \leq d$. Consider the following:

> **Recall:** *(Theorem) Rules of Inequality*
> Let $a, b, c \in \mathbb{R}$. The following statments must hold:
>
> - $(a > b) \wedge (b > c) \implies a > c$
> - $(a > b) \implies (a + c) > (a + b)$
> - $(a > b) \wedge (c > 0) \implies ca > cb$
> - $(a > b) \wedge (c < 0) \implies ca < cb$

Applying the theorem above, consider the following:

$$\begin{aligned}
&a < b \\
&a + c < b + c \\
&b + c \leq b + d \\
&a + c < b + c \leq b + d \\
&a + c < b + d
\end{aligned}$$

Part (b): If $0 < a < b$ and $0 \leq c \leq d$, prove that $0 \leq ac \leq bd$

Suppose $0 < a < b$ and $0 \leq c \leq d$

> **Recall:** *(Theorem) Rules of Inequality*
> Let $a, b, c \in \mathbb{R}$. The following statments must hold:
>
> - $(a > b) \wedge (b > c) \implies a > c$
> - $(a > b) \implies (a + c) > (a + b)$
> - $(a > b) \wedge (c > 0) \implies ca > cb$
> - $(a > b) \wedge (c < 0) \implies ca < cb$
>
> Ordering If $a - b \in P$, we write $a > b$ (or $b < a$).
> If $a - b \in P \cup \{0\}$, we write $a \geq b$ (or $b \leq a$).
> For any two real numbers a, b exactly one of the following holds:
>
> $$a > b, a < b, a = b$$

> **Recall:** *Properties of Positive Sets*
> Let $P \neq \emptyset, P \subseteq \mathbb{R}$ be called the set of positive real numbers.
> 1. If $a, b \in P$ then $a + b \in P$.
> 2. If $a, b \in P$ then $a \cdot b \in P$.
> 3. *Trichotomy Property:* If $a \in \mathbb{R}$, then exactly one of the following must be true:
>
> $$a \in P, \quad a = 0, \quad -a \in P$$

Applying the theorems above, observe the following:

$$(0 < a) \wedge (0 \leq c) \implies (0 \leq ac)$$

Thus the following statements hold:

$$0 \le c \le d$$
$$0 \le bc \le bd$$
$$0 < a < b$$
$$0 \le ac \le bc \le db$$
$$0 \le ac \le db$$

Question 2: Section 2.1-11

Part (a): If $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$

Suppose $a > 0$ and $\frac{1}{a} \not> 0$. Consider the following:

$$\frac{1}{a} \not> 0$$
$$(\frac{1}{a}) \cdot a \not> 0 \cdot a$$
$$(\frac{1}{a}) \cdot a \not> 0$$
$$1 \not> 0$$

Clearly $1 > 0$ thus this is a contradiction.
Thus we must accept the original proposition: $a > 0 \implies \frac{1}{a} > 0$.

Prove $\frac{1}{1/a} = a$:

$$\frac{1}{a} \cdot a = 1 \qquad \text{Multiplicative Inverse}$$

$$\frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \qquad \text{Multiplicative Inverse}$$

$$a \cdot \frac{1}{a} \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \qquad \text{Multipy a on both sides}$$

$$1 \cdot \frac{1}{\frac{1}{a}} = 1 \cdot a \qquad \text{Multiplicative Inverse}$$

$$\frac{1}{\frac{1}{a}} = a \qquad \text{Multiplicative Identity}$$

Thus we can say $\frac{1}{1/a} = a$

Now we have proven if $(a > 0)$, then $(1/a > 0)$ and $\frac{1}{1/a} = a$.

Part (b): If $(a < b)$, then $a < \frac{1}{2}(a+b) < b$

Consider the following:

$$a = a$$

$$\frac{1}{2}a = \frac{1}{2}a$$

$$\frac{1}{2}a + \frac{1}{2}a = \frac{1}{2}a + \frac{1}{2}a$$

$$\frac{1}{2}(a+a) = \frac{1}{2}(a+a)$$

$$\frac{1}{2}(2a) = \frac{1}{2}(a+a)$$

$$\frac{1}{2}2(a) = \frac{1}{2}(a+a)$$

$$a = \frac{1}{2}(a+a)$$

$$a < b \implies a + a < a + b$$

$$a < b \implies a + b < b + b$$

$$a < \frac{1}{2}(a+b) < b$$

Question 3: Sec 2.1-13

If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ iff $a = 0$ and $b = 0$.

Case 1: $\rightarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a^2 + b^2 = 0$ implies $a = 0$ and $b = 0$.
Consider the contrapositive statement, $a \neq 0$ or $b \neq 0$ implies $a^2 + b^2 \neq 0$:
Observe the following:

If $a \in \mathbb{P}$

$$a \in \mathbb{P} \implies a^2 = a \cdot a \in \mathbb{P} \implies a^2 > 0$$

If $-a \in \mathbb{P}$: (Consider $\forall\ n \in \mathbb{R}, (n - n) \cdot (-n) = 0$)

$$\begin{aligned} n\cdot(-n)+(-n)(-n)&=0\\ (-n)+(-n)(-n)&=0\\ \implies (-n)(-n)&=n^2\in\mathbb{P}\end{aligned}$$

Therefore, $-a^2 \in \mathbb{P}$. The same logic can be applied to b.

Observe If $a^2, b^2 \in \mathbb{P} \implies a^2+b^2 \in \mathbb{P}$.

Therefore we have proven the contrapositive statement if $a \neq 0$ or $b \neq 0$ implies $a^2+b^2 \neq 0$

Case 2: $\leftarrow$

Prove: If $a, b \in \mathbb{R}$, show that $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$.
Consider the following:

$$0\cdot 0 = 0 \implies \text{If } x = 0, \text{ then } x^2 = x\cdot x = 0\cdot 0 = 0$$

As $a = b = 0$ then naturally if $a = 0$ and $b = 0$ implies $a^2 + b^2 = 0$

Question 4: Section 2.1-14

If $0 \leq a < b$, show that $a^2 \leq ab < b^2$. Show by example that it does **not** follow that $a^2 < ab < b^2$.
Consider the following cases:

$$\begin{cases}\text{Case 1: } (a=0) \quad a<b \implies a^2 = ab < b^2\\ \text{Case 2: } (a>0) \quad a<b \implies (a^2<ab)\wedge(ab<b^2) \implies a^2<ab<b^2\end{cases}$$

Note the following:

$$(a^2 < ab < b^2) \wedge (a^2 = ab < b^2) \equiv a^2 \leq ab < b^2 \text{ which is what we wanted to show}$$

Likewise since if $a = 0 \implies a^2 = ab < b^2$ we can not satify the inequality $a^2 < ab < b^2$

Question 5: Section 2.1-20

Part a: If $0 < c < 1$, show that $0 < c^2 < c < 1$.

We can define $c < 1$ as $c = 1 - \epsilon$, where $\epsilon = \frac{1}{n}$ for some $n \in \mathbb{N}$ such that $0 < \epsilon < 1$.
Observe the following:

$$\begin{aligned} 1 &> 1-\epsilon\\ c &= 1-\frac{1}{n}\\ c^2 &= \left(1-\frac{1}{n}\right)^2 = 1-\frac{2}{n}+\frac{1}{n^2}\end{aligned}$$

We want to compare c and c^2. Since $n \geq 1$ and $\epsilon > 0$, multiplying the inequality $c < 1$ by c (where $c > 0$) preserves the sign:

$$\begin{aligned} c \cdot c &< 1 \cdot c \\ c^2 &< c \end{aligned}$$

Thus, combining with our initial range:

$$0 < c^2 < c < 1$$

Part b: If $1 < c$, show that $1 < c < c^2$.

We can define $c > 1$ as $c = 1 + \epsilon$, where $\epsilon = \frac{1}{n}$ for some $n \in \mathbb{N}$.

Observe the following:

$$\begin{aligned} 1 &< 1 + \epsilon \\ c &= 1 + \epsilon \\ c^2 &= (1 + \epsilon)^2 \\ &= 1 + 2\epsilon + \epsilon^2 \end{aligned}$$

Since $\epsilon > 0$, it must be that $\epsilon < 2\epsilon + \epsilon^2$. Therefore:

$$\begin{aligned} 1 < 1 + \epsilon &< 1 + 2\epsilon + \epsilon^2 \\ 1 < c &< c^2 \end{aligned}$$

Since we defined $\epsilon > 0$, the statement $1 < c < c^2$ holds.

Question 6: Section 2.1-23

If $a > 0$, $b > 0$, and $n \in \mathbb{N}$, show that $a < b$ iff $a^n < b^n$ (induct.)

$\forall\ a, b \in \mathbb{R}, a > 0, b > 0$ and $n \in \mathbb{N}$.
Consider $P(n) : a^n < b^n \iff a < b$
Observe $P(1)$:

$$\begin{aligned} a^1 < b^1 &\implies a < b \\ a < b &\implies a^1 < b^1 \end{aligned}$$

Suppose $P(k), k \in \mathbb{N}$:

$$a^k < b^k \implies a < b$$

Observe the following:

$$\begin{aligned} a^k &< b^k \\ a \cdot a^k &< b \cdot a^k \quad \text{Eq 1} \\ b \cdot a^k &< b \cdot b^k \quad \text{Eq 2} \end{aligned}$$

From Eq (1,2), we can derive the following

$$\begin{array}{rc} & a^{k+1} < b \cdot a^k \\ + & b \cdot a^k < b^{k+1} \\ \hline & a^{k+1} + (ba^k) < (ba^k) + b^{k+1} \end{array}$$

Thus we can apply algera to derive the final statement.

$$a^{k+1} + (ba^k) + (-ba^k) < (ba^k) + (-ba^k) + b^{k+1}$$
$$a^{k+1} < b^{k+1}$$

Thus we conclude $a^k < b^k \implies a^{k+1} < b^{k+1}$. Therefore by PMI $a^n < b^n$.
Observe application of Trichotomy Property (Note we concluded: $a^n < b^n$).

$$\begin{cases} \text{Cases 1: } (a = b) & \implies a^n = b^n \lightning \\ \text{Cases 2: } (a > b) & \implies a^n > b^n \lightning \\ \text{Cases 3: } (a < b) & \text{Must hold by Trichotomy Property} \end{cases}$$

Therefore, strict inequality implies strict inequality in reverse.
Therefore, we have proven both direction.
We can state, $a < b$ iff $a^n < b^n$

Question 7: Section 2.1-25

Assuming the existence of roots, show that $c > 1 \implies c^{1/m} < c^{1/n}$ iff $m > n$

Lemma 1: Integer Powers of $c > 1$

First, we establish that for base $c > 1$, larger integer exponents yield larger values. We show $c^b > c^a \iff b > a$ for $a, b \in \mathbb{N}$.

Forward ($b > a \implies c^b > c^a$): Let $b > a$. Then $b - a = k \in \mathbb{N}$ where $k \geq 1$. Using Bernoulli's Inequality for $c = 1 + x$ where $x > 0$:

$$c^k = (1 + x)^k \geq 1 + kx > 1$$

Multiplying both sides by c^a (which is positive):

$$c^k \cdot c^a > 1 \cdot c^a \implies c^{a+k} > c^a \implies c^b > c^a$$

Reverse ($c^b > c^a \implies b > a$): We proceed by contradiction. Assume $c^b > c^a$ but $b \leq a$.

$$\begin{aligned} \text{If } b = a &\implies c^b = c^a \quad \text{(Contradiction)} \\ \text{If } b < a &\implies a - b = k \in \mathbb{N}, k \geq 1 \\ &\implies c^a > c^b \quad \text{(By Forward proof)} \\ &\text{(Contradiction)} \therefore c^b > c^a \iff b > a \end{aligned}$$

Lemma 2: Roots of $c > 1$ exceed 1

Let $c > 1$. Let $y = c^{1/k}$ for some $k \in \mathbb{N}$. We claim $y > 1$.

Proof. Assume for contradiction that $y \leq 1$. Since $y > 0$, raising both sides to the power of k:

$$y \leq 1 \implies y^k \leq 1^k \implies c \leq 1$$

This contradicts the premise $c > 1$. Therefore, $c^{1/k} > 1$. □

Main Proof

Let $N = mn$. Let $b = c^{1/N} = c^{1/mn}$. From **Lemma 2**, we know $b > 1$. We can rewrite the terms $c^{1/m}$ and $c^{1/n}$ using base b:

$$c^{1/m} = (c^{\frac{1}{mn}})^n = b^n$$
$$c^{1/n} = (c^{\frac{1}{mn}})^m = b^m$$

We wish to show $c^{1/m} < c^{1/n} \iff m > n$. Substituting the expressions in terms of b:

$$b^n < b^m \iff m > n$$

Since $b > 1$, we apply **Lemma 1**:

$$b^m > b^n \iff m > n$$

This is exactly the statement we needed to show.

Question 9: Section 2.2-1

If $a, b \in \mathbb{R}$ and $b \neq 0$ show the following:

Part (a): $a, b \in \mathbb{R} \implies |a| = \sqrt{a^2}$

$$\text{Recall: } |a|^2 = |a| \cdot |a| = |aa| = |a^2| = a^2 \implies$$
$$|a|^2 = a^2$$
$$\sqrt{|a|^2} = \sqrt{a^2}$$
$$|a| = \sqrt{a^2}$$

Part (b): $a, b \in \mathbb{R} \implies |a/b| = |a|/|b|$

$$\begin{cases} \text{Case 1: } b > 0 & |a/b| = |a| \cdot |1/b| \implies 1/b > 0 \\ \text{Case 2: } b < 0 & |a/b| = a \cdot -(1/b) = a \cdot 1/(-b) \implies 1/b < 0 \end{cases}$$
$$\therefore |a/b| = |a|/|b|$$

Question 10: Section 2.2-2

Prove: $a, b \in \mathbb{R} \implies |a + b| = |a| + |b| \iff ab \geq 0$

Note all statements are reversible.

$$\iff |a + b| = |a| + |b|$$
$$\iff |a + b|^2 = (|a| + |b|)^2$$
$$\iff (a + b)^2 = |a|^2 + 2|ab| + |b|^2$$
$$\iff a^2 + 2ab + b^2 = |a|^2 + 2|ab| + |b|^2$$
$$\iff a^2 - (a^2) + 2ab + b^2 - (b^2) = |a|^2 - (a^2) + 2|ab| + |b|^2 - (b^2)$$
$$\iff 2ab = 2|ab|$$
$$\iff ab = |ab|$$

Note definition of absolute value invokes $\forall\ x \in \mathbb{R}, x = |x| \iff x \geq 0$. Therefore $ab \geq 0$
We can conclude that the statement $a, b \in \mathbb{R} \implies |a + b| = |a| + |b| \iff ab \geq 0$ holds.

Question 11: Section 2.2-12

Find all $x \in \mathbb{R}$ that satisfy the inequality $4 < |x+2| + |x-1| < 5$

Consider the following:

$$\begin{aligned} &f(x) := |x+2| + |x-1| \\ &f(0) = 3 \\ &f(1) = 3 \quad f(-1) = 3 \\ &f(2) = 5 \quad f(-2) = 3 \\ &f(3) = 7 \quad f(-3) = 5 \end{aligned}$$

> By inspection I suspect there are three cases. We can construct these cases to encompass all of $\mathbb{R}$.
>
> $$\begin{cases} (x \geq 1) \\ (-2 < x < 1) \\ (x \leq -2) \end{cases}$$

Case 1: $(x \geq 1)$

$$\begin{aligned} &|x+2| = (x+2) = x+2 \\ &|x-1| = (x-1) = x+1 \\ &\implies 4 < (x+2) + (x-1) < 5 \\ &\implies 4 < 2x+1 < 5 \\ &\implies 3 < 2x < 4 \\ &\implies \frac{3}{2} < x < 2 \end{aligned}$$

Case 2: $(-2 < x < 1)$

$$\begin{aligned} &|x+2| = (x+2) = x+2 \\ &|x-1| = -(x-1) = -x+1 \\ &\implies 4 < (x+2) + (-x+1) < 5 \\ &\implies 4 < 3 < 5 \\ &\quad \text{(Impossible)} \end{aligned}$$

Case 3: $(x \leq -2)$

$$\begin{aligned} &|x+2| = -(x+2) = -x-2 \\ &|x-1| = -(x-1) = -x+1 \\ &\implies 4 < (-x-2) + (-x+1) < 5 \\ &\implies 4 < -2x-1 < 5 \\ &\implies 5 < -2x < 6 \\ &\implies -\frac{5}{2} > x > -3 \end{aligned}$$

Observe there is no way to satisfy the inequality in (**Case 2**). Therefore we can conclude possible solutions may only exist in **Case 1,3**. Thus are bounded region with solutions to the inequality are as follows:

$$(-\frac{5}{2} > x > -3) \wedge (\frac{3}{2} < x < 2)$$

Question 12: Section 2.2-17

Show that if $a, b \in \mathbb{R}$ and $a \neq b$ then $\exists$ ϵ-neighborhoods U of a and V of b $\ni U \cap V = \emptyset$

$$\begin{aligned}
U_\epsilon(a) :&= \{u \in \mathbb{R} : |u - a| < \epsilon\} \\
&= \{u \in \mathbb{R} : -\epsilon < u - a < \epsilon\} \\
&= \{u \in \mathbb{R} : a - \epsilon < u < \epsilon + a\} \\
&= (a - \epsilon, a + \epsilon) \\
V_\epsilon(b) :&= \{v \in \mathbb{R} : |v - b| < \epsilon\} \\
&= \{v \in \mathbb{R} : -\epsilon < v - b < \epsilon\} \\
&= \{v \in \mathbb{R} : b - \epsilon < v < \epsilon + b\} \\
&= (b - \epsilon, b + \epsilon)
\end{aligned}$$

Observe $U \cap V$:

$$\begin{aligned}
U \cap V &:= \{x \in \mathbb{R} : (|x - a| < \epsilon) \wedge (|x - b| < \epsilon)\} \\
&\implies \forall\ \epsilon < \frac{1}{2}|a - b|, U \cap V = \emptyset
\end{aligned}$$